

Candidate Profile of Abinaya Shanmugam

Abinaya communicates well at interview and projects herself as confident and professional. She is a natural and effective communicator and is able to clearly articulate her skills and experiences.

Abinaya is an experienced Full Stack Java Developer with 12 years of hands-on engineering expertise and a track record of leading complex development initiatives. She has had some significant recent experiences where she is working across the full stack.

She is currently working for Earth Sciences New Zealand [Earth Sciences New Zealand | Earth Sciences New Zealand](#) and has taken a key role in the rebuild of their legacy Oracle Forms and Struts applications into a high-performance web platform using Java 21, Spring Boot, and Thymeleaf, significantly improving maintainability, scalability, and user experience.

She offers deep capability across microservices architecture, designing and implementing robust REST and SOAP services using Spring Boot, Hibernate, and PostgreSQL. Her recent work includes architecting and executing the strategic migration from Oracle DB to MongoDB, designing efficient NoSQL schemas to support future scalability and performance.

At Earth Sciences she is required to cover a range of tasks and is responsible for the design components as it relates to her project and adheres to all security related requirements. They don't have a Software Testing team and therefore their Developers are required to manage testing related to their work.

Salary expectations: \$110,000 - \$120,000 per annum

Availability: 4 week notice period

Abinaya Shanmugam

📍 Auckland , New Zealand

Experience Summary

Senior Software Engineer with 12+ years of experience designing, developing, and delivering enterprise-scale applications across Telecommunications, Climate Science, Insurance, Finance, and Retail domains. Strong expertise in Java, Spring Boot, Microservices, REST APIs, Cloud Technologies (AWS/Azure), and Distributed Systems. Experienced in solution design, stakeholder engagement, technical leadership, mentoring teams, and delivering high-quality software throughout the full Software Development Lifecycle. Passionate about modern engineering practices including cloud-native development, DevOps, containerization, and continuous improvement.

Key Skills

- **Domain Knowledge:** Climate Science, Insurance, Finance, Retail, Telecommunications.
- **Backend Technologies:** Java (21), Spring Boot, Microservices, REST APIs, Distributed Systems
- **Web Technologies:** HTML, JavaScript, Thymeleaf, AngularJS, JSON, Pig Script.
- **Event-Driven & Integration:** Apache Kafka, API Gateway, Asynchronous Processing, Event-Driven Architecture
- **Frameworks & Tools:** Spring MVC, Spring Boot, Hadoop, Hibernate ORM, JDBC, Camunda, OpenShift, Jenkins, Maven, Swagger/Open API
- **Cloud & Platforms Technology:** Azure, AWS (S3, IAM, EC2, RDS, API Gateway, Lambda, EKS), Docker, CI/CD
- **Database:** MongoDB, Oracle, MYSQL, PostgreSQL, NoSQL.
- **Version Control System:** GIT, Bit Bucket.

Professional Experience

**Earth Sciences New Zealand
Auckland, NZ**

**Software Engineer
April 2025 – Present (Contract)**

- Designed and developed scalable full-stack applications using Java 21, Spring Boot, REST APIs, and responsive web technologies to support mission-critical climate and weather data processing platforms.
- Spearheaded the modernization of legacy Oracle Forms and Struts applications into cloud-ready Spring Boot microservices with modern web-based user interfaces, improving maintainability, scalability, performance, and user experience. Contributed to the initial phase of migrating Oracle Database applications to MongoDB to enhance scalability and performance.
- Driving the strategic migration of Oracle Database architecture to MongoDB to improve system scalability, data performance, and long-term maintainability.
- Developed and enhanced frontend components using AngularJS, Thymeleaf, HTML, CSS, JavaScript, and Bootstrap, integrating them seamlessly with Spring Boot REST APIs to deliver responsive and user-friendly applications.
- Partnered with stakeholders, product owners, architects, and cross-functional teams to define business requirements, estimate delivery effort, design scalable solutions, and participate in solution architecture reviews and cloud adoption initiatives.
- Engineered high-performance microservices to consume real-time MetService data, enabling efficient APIs and on-demand alerting systems for users and fire stations.
- Designed, containerized, and deployed cloud-native applications using Docker and Kubernetes (AWS EKS), enabling scalable, resilient, and highly available production workloads.
- Contributed to CI/CD pipelines for automated build, testing, deployment, and release processes, improving software delivery speed, quality, and deployment reliability.
- Implemented secure authentication and authorization using OAuth2/OIDC and role-based access control (RBAC) to protect enterprise applications, REST APIs, and frontend applications.
- Collaborated closely with UI/UX teams and backend engineers to deliver end-to-end features, ensuring seamless integration between frontend interfaces and backend services.

- Ensured application reliability and operational stability through proactive monitoring, production support, troubleshooting, and resolution of critical production incidents.
- Managed the complete Software Development Lifecycle (SDLC), including requirements gathering, technical design, frontend and backend development, testing, deployment, production support, and continuous improvement using Agile methodologies.

**Spark New Zealand
Auckland, NZ**

**Java Developer
July 2022 – January 2024 (Contract)**

- Designed and developed REST APIs, distributed microservices, and Kafka-based event-driven integrations supporting critical business workflows and enterprise systems
- Translated legacy Velocity scripts into RESTful web services to integrate necessary logic and functionalities.
- Efficiently resolved SIT/UAT bugs within tight timelines.
- Utilized Camunda BPMN and Swagger open-source tools.
- Worked directly with business stakeholders and customers to understand requirements, manage expectations, and deliver scalable enterprise solutions.
- Involved in all phases of development and testing, including UT, SIT, and Pre-Production.
- Implemented suitable Git strategies for effective version control of the code.
- Solely responsible for automating a longstanding data gathering process, involving extraction from various systems and updating details in the Spark databases.
- Developed Stored Procedures to streamline the necessary app functions for automation and scheduled them as CRON jobs.
- Engaged directly with clients throughout the project and implemented a mobile insurance module tailored to specific requirements.
- Utilized OpenShift to containerize and deploy applications on Azure to improve security and scalability.

Bangalore, Infosys

March 2018 – June 2022

Walmart

Java/MDM Developer

April 2020 – June 2022

- Understood client needs and devised code logic to meet them.
- Wrote Business conditions and Business rules with Graphs and Schedulers to implement logic.
- Designed and developed custom workflow processes including initiation, user approval, maintenance, and completion.
- Created and maintained various entities and handled configuration related to export profiles.
- Performed Production deployment and fixed critical production bugs on time.
- Used API calls while implementing Business rules for specific requirements.
- Expertise in different domain models used in Riversand/Syndigo.
- Led and mentored a team of 6–7 engineers, conducting code reviews, driving technical decisions, ensuring delivery quality, and promoting engineering best practices.
- Played the Scrum Master role and was involved in all Agile activities from Sprint planning to Sprint Closure.

VISA

Java Developer

Bangalore, Infosys

March 2018 – March 2020

- Involved in the project from the estimation phase through to the PROD deployment.
- Translated client requirements into User Stories and assigned them to the team for development.
- Developed optimized, high-quality code and performed peer reviews.
- Used code coverage and code quality tools during development.
- Performed unit testing for each functionality and managed build and deployment of components.
- Mentored and led the team by providing necessary technical input and help.

MetLife

Java Developer

Coimbatore, Cognizant Technology Solutions

July 2015 – February 2018

- Redesigned a legacy PowerBuilder application into a Java-based web application.
- Took ownership of a module and developed it fully based on client requirements.
- Performed Unit Testing and Integration Testing for the created module.
- Maintained all technical documents for stakeholder sign-off.
- Worked in all phases of SDLC, from requirement gathering to PROD deployment.

Dun & Bradstreet
Coimbatore, Cognizant Technology Solutions

Backend Developer
November 2013 – July 2015

- Analysed, designed, and developed application modules from scratch.
 - Upskilled in Hadoop in a short span of time and contributed to the project.
 - Analysed mainframe code components and converted them to PIG Scripting code.
 - Created Hadoop clusters, data nodes, and worked on configuration.
 - Refactored code which saved maintenance efforts effectively.
-

Education

Bachelor's degree in information technology Anna University, Chennai, India |
June 2009 - June 2013